

UnlimitedCode Agentic AI Bootcamp

Week 5 Assignment

Building a University Student Services AI Agent

Developer Track

Topic: Retrieval-Augmented Generation (RAG), Vector Search, LangGraph Retrieval Workflows

Suggested Due Date: One week from assignment release

Assignment Overview

You have been hired by a university to build an AI assistant that answers student questions using institutional knowledge. The assistant must retrieve information from university documents, ground its answers in retrieved evidence, and avoid hallucinations.

Knowledge Sources

Student Handbook

Academic policies, grading system, attendance requirements, graduation requirements, student code of conduct.

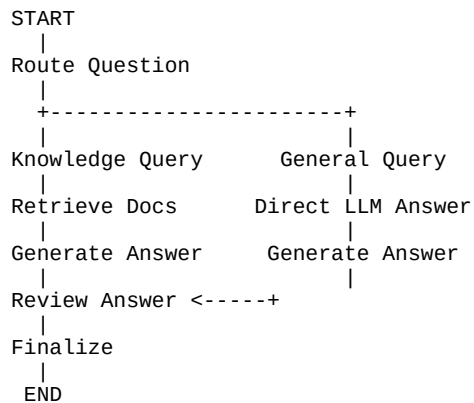
Tuition & Fees Guide

Tuition costs, payment deadlines, scholarships, refund policies.

Campus Services Guide

Housing information, library services, IT support, career services, student health center.

Required Architecture



Functional Requirements

- Load at least three documents.
- Chunk documents using RecursiveCharacterTextSplitter.
- Create embeddings using OpenAI embeddings.
- Store vectors in FAISS or Chroma.
- Build a retriever with k=5.
- Implement a LangGraph workflow.
- Add conditional routing.
- Generate grounded answers only.
- Implement a review node.
- Handle missing information gracefully.

Implementation Examples

Chunking Example

```
text_splitter = RecursiveCharacterTextSplitter(  
    chunk_size=1000,  
    chunk_overlap=200  
)
```

Retriever Example

```
retriever = vector_store.as_retriever(  
    search_kwargs={"k": 5}  
)
```

Metadata Example

```
{  
    "source": "student_handbook.pdf",  
    "section": "attendance_policy"  
}
```

Routing Examples

```
Knowledge Questions  
-----  
What GPA do I need to graduate?  
What is the tuition refund policy?  
Where can I get IT support?  
  
General Questions  
-----  
Tell me a joke.  
What is Python?  
Who won the World Cup?
```

Bonus Challenges

Bonus Challenge 1 (+10 points)

Add conversational memory using MessagesState or MemorySaver.

Bonus Challenge 2 (+15 points)

Build a multi-source agent capable of routing between: University Documents (RAG) Weather Tool Web Search Tool

Deliverables

1. GitHub Repository
2. Architecture Diagram
3. README including: Setup Instructions Graph Workflow Screenshots Example Questions

AI Fair Use Policy

AI tools such as ChatGPT, Claude, Gemini, Copilot, Cursor, and similar assistants may be used during this assignment. However:

- You must understand all code submitted.
- You must be able to explain every major design decision.
- Blind copy-paste without understanding is prohibited.
- All AI assistance should be acknowledged in your README.
- During code review, learners may be asked to explain any part of their solution.

Grading Rubric (100 Points)

Category	Points
Document Loading	10
Chunking Strategy	10
Embeddings & Vector Store	15
Retriever Quality	15
LangGraph Workflow	15
Conditional Routing	10
Review Node	10
Grounding & Hallucination Prevention	10
Code Quality	5
README & Documentation	10

Bonus: +25 points available for memory and multi-source routing.